

MBARI Waypoint Management System

Requirements Document

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Draft 4: 10/25/1999

Introduction

This document describes requirements for the MBARI Waypoint Database and the software applications for loading data into it and extracting data from it.

The database and the processes described here were developed to:

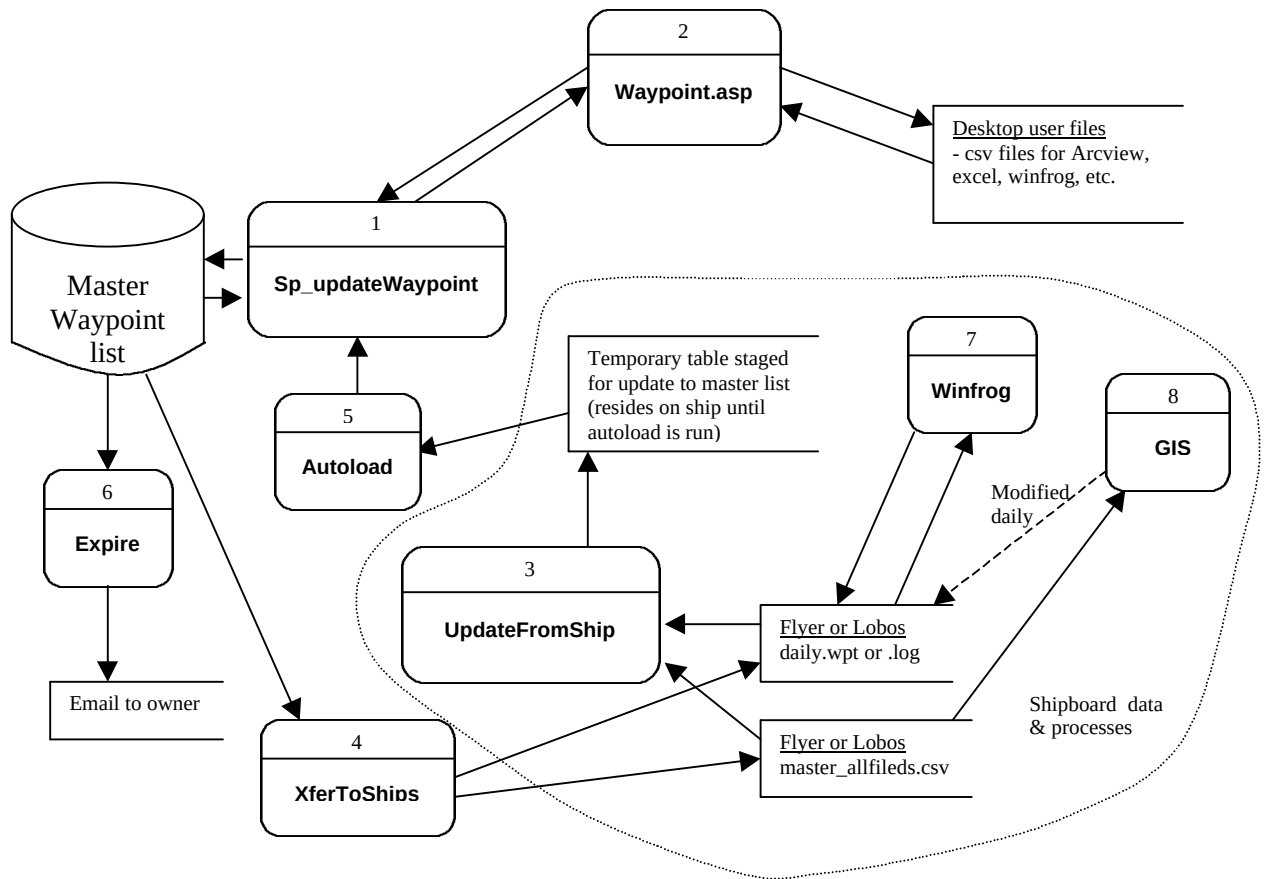
- Maintain a "master" list of waypoint names that can be used by both MBARI vessels for ROV operations.
- Provide mechanisms to update this list following shipboard activities.
- Allow for editing, insertion, and deletion of waypoints from the scientist's desktop.
- Provide a way to search and extract waypoints for import into other programs

The system is coordinated with MBARI's Expedition Database pre-cruise planning and post-cruise reporting system. The waypoints are for navigational use and may include midwater, benthic, and shore locations.

The system is intended to serve the institutional needs of MBARI and may become a core datastream. It is needed now to facilitate communication, prevent confusion, and avoid any misunderstandings of waypoint names or locations.

Data Flow Diagram

A data flow diagram summarizes the waypoint management system. It shows the flow of data between data sources (boxes missing an edge) and processes (rounded boxes). Short descriptions of the processes are given below. These processes need to be developed by the Information Applications group.



Process Descriptions

1. **Sp_updateWaypoints**

is a stored procedure within the database which ensures consistency by warning of duplicate names or locations. It issues warnings or errors to the application programs that use it for inserting or updating waypoints to the master list.

2. **Waypoint.asp**

is an Active Server Page that provides the web interface to the master waypoint list.

This program will provide the following:

- search capability
- edit, add, or delete specific waypoints via html form interface
- export a selected list of waypoints to a comma separated value file that an end user can then import into another application such as ArcView.
- import a set of new waypoints from a csv file on the users workstation

3. **UpdatesFromShip**

is a process that reconciles the new (or edited) waypoints entered (or changed) during ROV operations. It is a program that runs automatically following operations. It creates a temporary table of new waypoints that is staged in a local file on a shipboard computer. This staging is required so that the interactive process is not held hostage to a missing network connection to shore.

4. **XferToShips**

is a process which runs a program that each morning copies at least 2 files to each ship: 1) a daily.wpt file which is a subset of waypoints from the master list. It is defined by the pre-cruise region description or a selection of waypoints based on distance from a target waypoint. The transferred daily.wpt file can be overwritten aboard ship by using ArcView tools to select a different subset from the master_allfields.csv file. 2) a master_allfields.csv file which contains all of the waypoints existing in the master list at the time of transfer.

5. **Autoload**

is a process that runs automatically at night to take the temporary table of new waypoints created by UpdatesFromShip.exe process and loads them into the master list via the Sp_updateWaypoints stored procedure.

6. **Expire**

is a process that is set to run at a regular frequency, such as weekly. It checks the expiration date against the current date and deletes the waypoint from the master list. An email containing all the waypoint information that was deleted is sent to the waypoint owner. The owner can re-enter the waypoint, if desired, through the waypoint.asp process.

7. Winfrog

is a navigation tool used by the ship and ROV operators. It reads and writes waypoints to and from waypoint (daily.wpt) and log event (daily.log) files.

8. GIS

includes navigational tools other than winfrog that may be used by scientists. Examples are ArcView with MBARI added extensions or the application Gnav. Both of these can read and write waypoint and log files, as well as a host of other kinds of files.

Data Dictionary

The fields for the master list of waypoints are defined as follows:

WaypointName	A short name for use in winfrog. This may be a code where the name is composed of owner's initials, an 8-character location name, and depth in meters. [Created from database fields] (15 characters - limit imposed by winfrog)
WaypointShortName	An 8-character location name (when combined with owner initials and depth, must be unique) (8 characters)
WaypointLongName	A descriptive longer unique name for the waypoint location.
Owner	The first and last name of the owner or originator (char 255)
UserGroup	e.g. Bio / geo / chem (char 255)
DateTimeOfCreation	Time of waypoint creation(DateTimeGroup)
Latitude	decimal degrees (float)
Longitude	decimal degrees (float)
LatitudeDM	degrees, decimal minutes, e.g. 'N36 32.1890' [Computed from Latitude] (string)
LongitudeDM	degrees, decimal minutes, e.g. 'W121 59.6390' [Computed from Longitude] (string)
Easting	[Computed from Latitude/Longitude] (float)
Northing	[Computed from Latitude/Longitude] (float)
UTMZone	UTM zone for Easting/Northing calculation (int)
RegionName	Name of region a waypoint resides in (Char 255)
WaypointDepth	Depth in meters (float)
Comment	Any string, keywords, etc. (Char 2048)
Status	A flag designating <i>temporary</i> or <i>permanent</i> (char 255)
Lifetime	days this waypoint is to exist before expiring (int)
HomerCode	Homer or Nav beacon code (char 255)
WaypointID	Autoincrementing identity column (int)

Constraints and Assumptions

The system must operate within the following environment and constraints:

1. An accurate pre-cruise plan containing the region of operation is filed before the day's operations. This will determine the contents of the daily files as well as the owner of the waypoints that are created during the operations.
2. All transfers of waypoints to and from ship will be mostly automatic, requiring minimal end-user interaction.
3. All waypoints entered into the master list following shipboard operations will have a status of temporary with a lifetime of 7 days.
4. The existing Expd database (SQL server) will be used for storing the master waypoint list.
5. The ship-to-shore and shore-to-ship transfer processes must tolerate network disconnects. The use of floppy disks and printed lists is an option.
6. This system is designed for navigational use in order to improve operational efficiency. Benefits to science are secondary.
7. Time of waypoint creation at the exact time of creation (within 5 seconds) must be recorded in some manner.